

NIC API Exchange Platform



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1. Introduction

Today, Governments at all levels are increasingly turning to Web and Mobile to improve their communications and responsiveness with the citizens by building e-Governance applications. These e-Governance applications mostly work in silos and are not integrated with e-Governance applications from other departments and private partners.

Application Programming Interface(API) can act as a gateway to e-Governance data and services by not only promoting software interoperability for e-Governance applications but at the same time provide secure and controlled access to data and services of e-Governance applications to stakeholders, including partners and citizens. APIs provide speedy application development, wider reach of relevant and updated data and content, automation and Integration with efficiency. Moreover, technologies - cloud, mobile, big data, and the Internet of Things (IoT) - demand a new level of connectivity that cannot be achieved with yesterday's integration approaches. To put in simple words APIs aims provides speedy application development, wider reach of relevant and updated data and content, Automation and Integration with efficiency.

An effective API Strategy will drive future direction for nationwide e-Governance data and services. It will empower digital transformation, opening the door to new types of solutions, drive innovation, improve time-to-value, and open up new possibilities for creative e-Governance models.

Centralization is the key – legacy backend systems do not need to be updated and APIs do not need to be designed with every potential client platform in mind . A well planned Centralized API Framework System providing complete API Lifecycle Management with following key components is desired:

- The core component of a full-featured API Management solution is an API Gateway. It act as an API front-end, receives API requests, enforces throttling and security policies

- A collection of tools Publishing Tools that API providers use to define APIs, for instance using the OpenAPI Specifications
- Community site i.e. Developer portal/API store that can encapsulate for API users in a single convenient source information and functionality including documentation, tutorials, sample code, software development kits, an interactive API console.
- Functionality to monitor API usage and load through Reporting and Analytics.

2. Purpose of the Document

With an increase in need of e-Governance applications to provide data and services to other government entities, partners and citizens, the use of APIs has increased many folds. Hence there is a need to have common platform where the APIs can published, discovered and used with mandatory features like security, and analytics.

This document gives an insight into the NIC Open API Management Solution from Publisher's perspective for creating and publishing an API and from developer's perspective for discovery, subscription and testing an API.

Publishers are the ones who publish the APIs on NIC Open API Management Solution to other government entities, partners and citizens. Using the solution, they can manage the whole life cycle of the published API.

Developers are the ones who consume the API in their applications. Using this solution one can access the data and government services on a centralised platform.

The document will help the user in understanding:

- API Life Cycle Management.
- API.
- Open API.
- Best practices to be followed for creating an API.
- NAPIX API Exchange Platform.
- Guide for using NAPIX API Exchange Platform -Developer Portal /Store

- To Discover an API.
- To Subscribe an API.
- To Invoke/Test an API.
- To Consume an API in the developer application.

3. API

An application program interface is a set of routines ,protocol,objects or functions that developers use to create the software or interact with the external system. In general it is set of clearly defined methods of communication between various software components. An API defines the correct way for a developer to request services from an operating system (OS) or other application and expose data within different contexts and across multiple channels. It lists a set of operations along with description of what the operations do so that developers can use those methods or operations. The developer doesn't need to know how exactly the operation is happening ,they just need to know what the inputs we have to give to the API and what outputs they will get in return. API allow developers to save their time by taking advantage of already available platform. This reduces the amount of code the developers have to create and also provides more consistency for the applications for the same platform.

APIs take three basic forms:

- Internal API,
- Open API and
- Partner API.

Internal API

Private APIs, or internal APIs, are published internally for use by the developers of the same organization to improve its own services. Private APIs are not exposed to third parties. Available only to NIC applications and users.

Open API

Public APIs, or open APIs, are published publicly and can be used by any third-party. There are no restrictions on these APIs. Available to non NIC applications and users Examples include the general public, other agencies, or private companies.

Partner API

Partner APIs can only be used by specific parties with whom the government agrees to share data. Partner APIs are used within other government entities and partners often to integrate software between the entities.

4. NIC API Exchange Platform

NIC API Exchange Platform delivers a single, centralised platform for users to design and build API's and publish it so that citizens can use for developing their own applications. Adopting NIC API Exchange Platform for APIs allows the users to easily design APIs quickly, integrate them with ease, collaborate effortlessly, and scale securely and reliably. With tools like swagger to support every step in the development and implementation of APIs, Bharat API Exchange Platform is an effective way to manage APIs from a single platform.

NIC API Exchange Platform features:

- It provides automated, model-driven tools for API creation (swagger).
- Provides traffic management and built-in security features for API and data protection.
- Self-service developer/store for empowering the developers.
- Provides API usage analytics for API providers as well as consumers.
- API testing and monitoring without coding.

NIC API Exchange Platform makes it easy to:

- Define and create the API
- Define policies like throttling criteria subscription tiers and token validity management.
- Publish the API.
- Test the API.
- Subscribe the API.
- Test the API
- Consume the API.

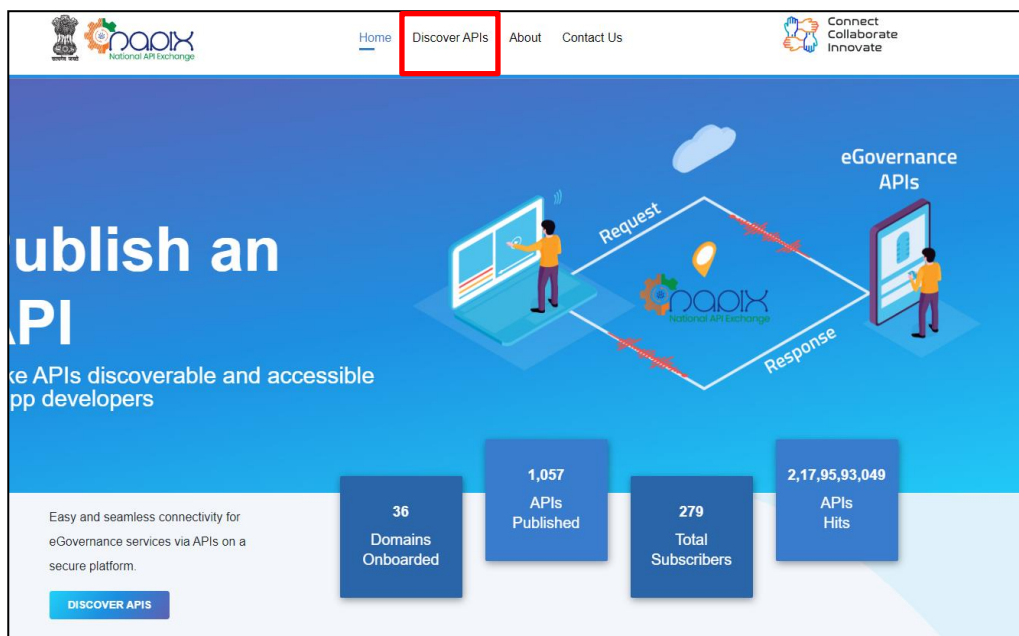
5. Guide for using NIC API Exchange Platform-Developer Portal/Store

The Step by procedure for a developer/Consumer includes

- Discover an API
- Subscribe an API
- Invoke/Test an API
- Consume the API in the end application

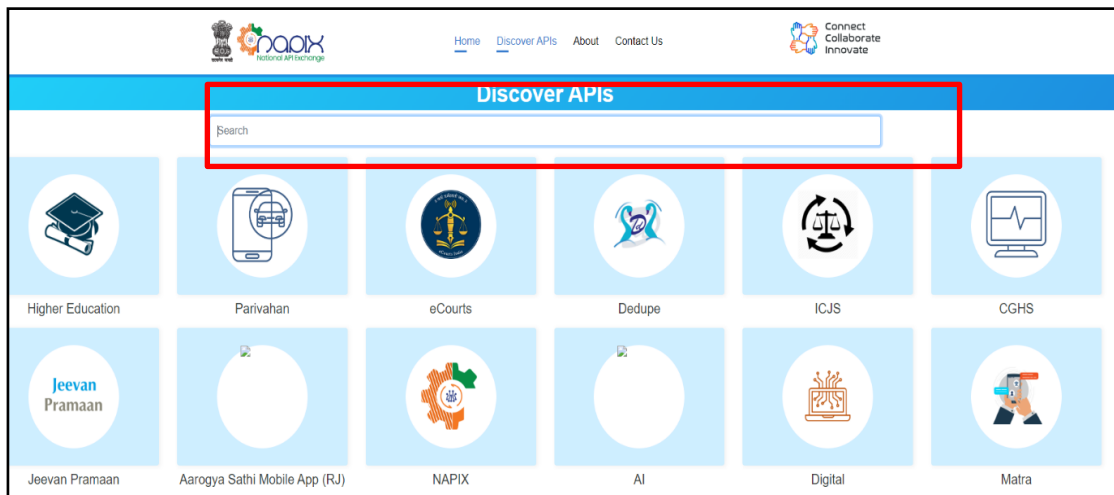
5.1 Discover an API

a) Go to <https://napix.gov.in>



b) Proceed to Discover APIS Menu

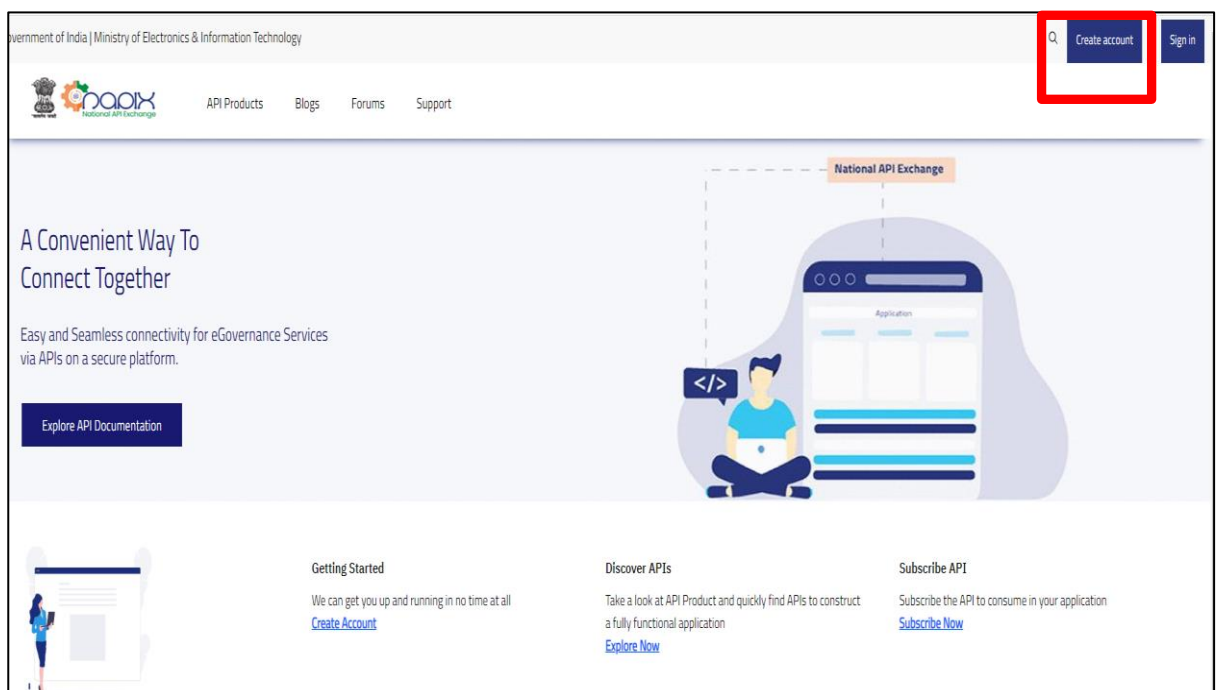
c) Search for the desired domain to find the API.



d) Go to desired domain for searching the APIs

5.2 Subscription of an API

a) Click on create Account to create a new account



b) Fill in the desired details

API Developer Portal

Sign up

Sign up with AI Catalog User Registry

Username *

Several special characters are allowed, including space, period (.), hyphen (-), apostrophe ('), underscore (_), and the @ sign.

Email address *

A valid email address. All emails from the system will be sent to this address. The email address is not made public and will only be used if you wish to receive a new password or wish to receive certain news or notifications by email.

First Name *

Last Name *

Consumer organization *

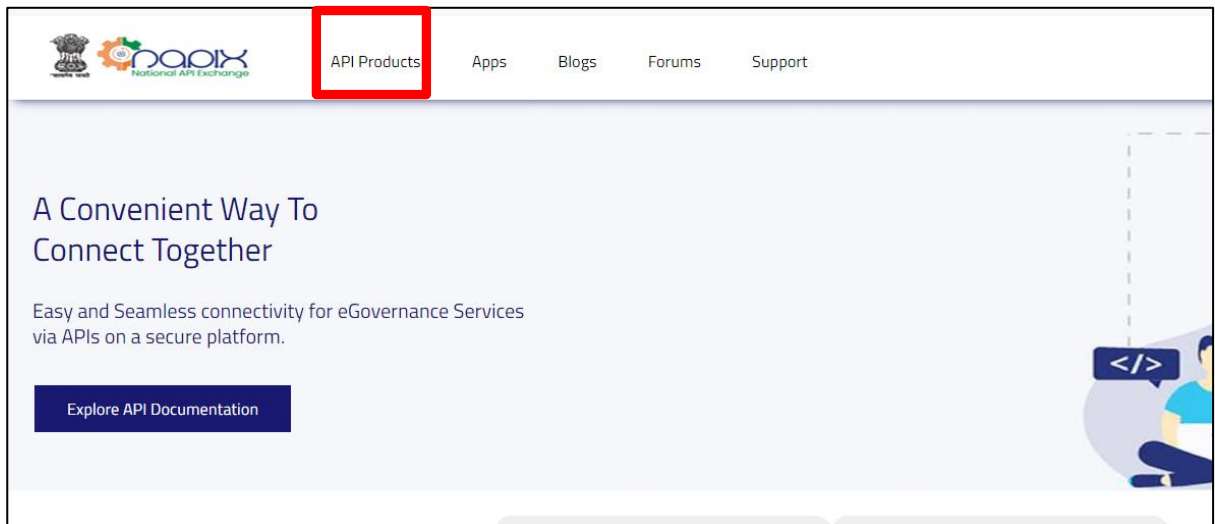
Provide a name for your consumer organization such as "ACME Enterprises".

Password *

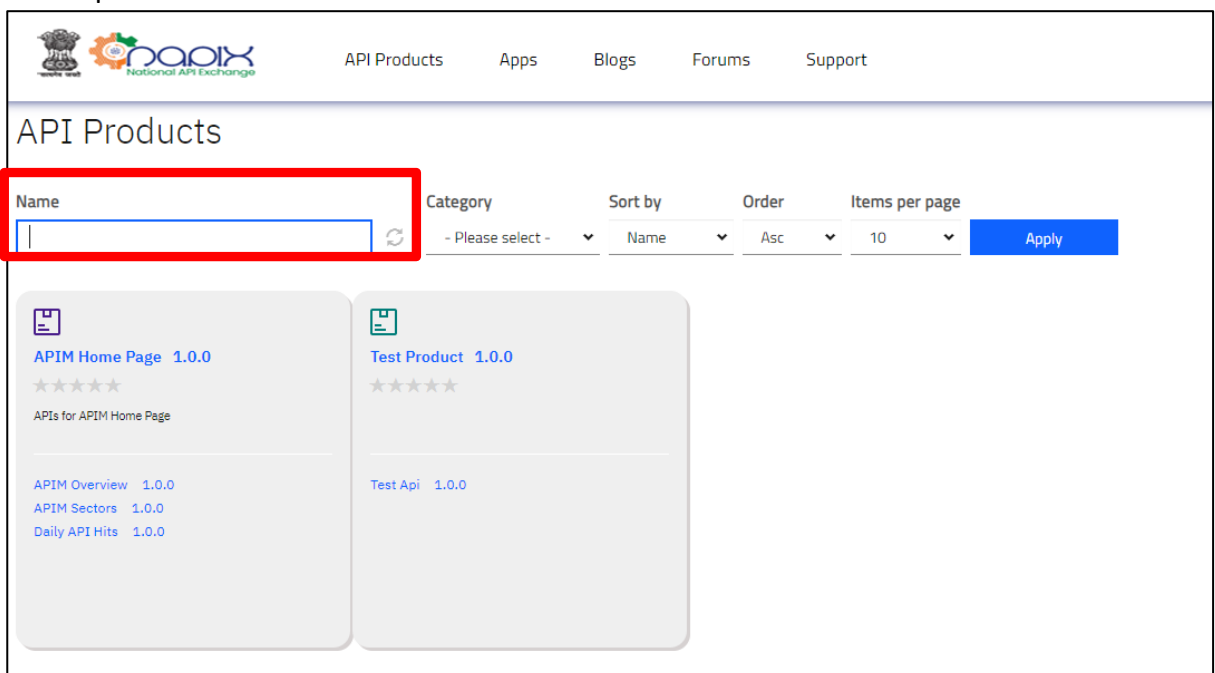


- c) Once the user fills the details the request will go for admin approval.
- d) Once the account gets approved by the API development team, the user will get an activation link over mail

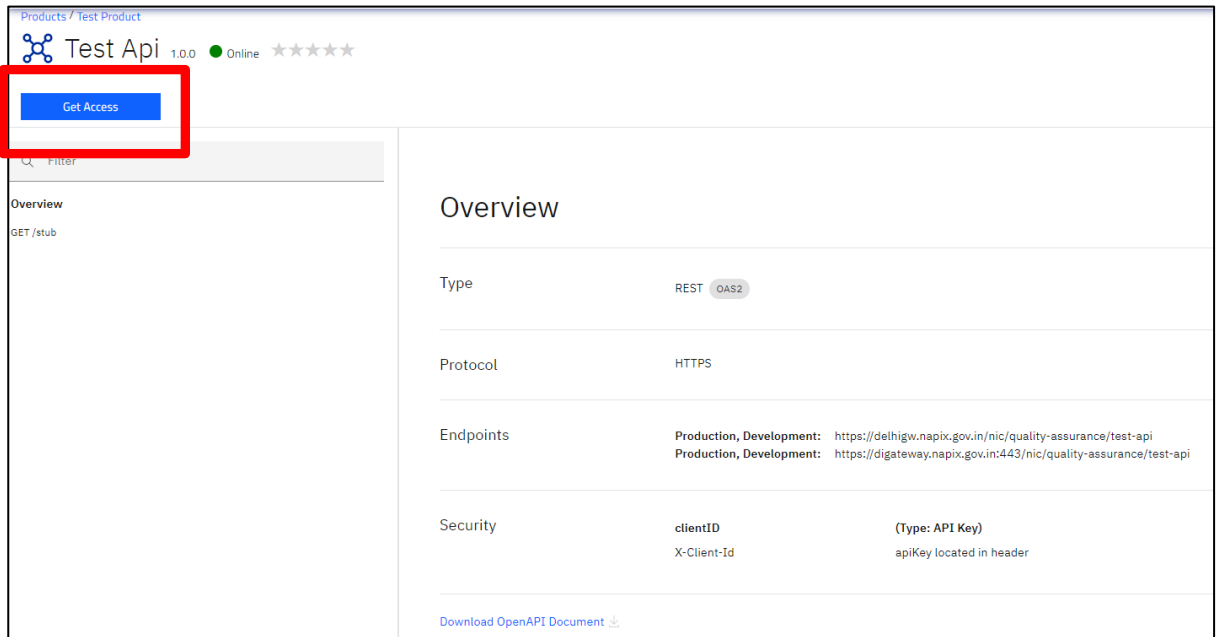
- e) Click on the activation link received in the mail and login with the credentials provided during the signup process (username and password).
- f) After successfully logs in to the portal, Click on API products section to find the available Products (APIs are bundled inside a product)



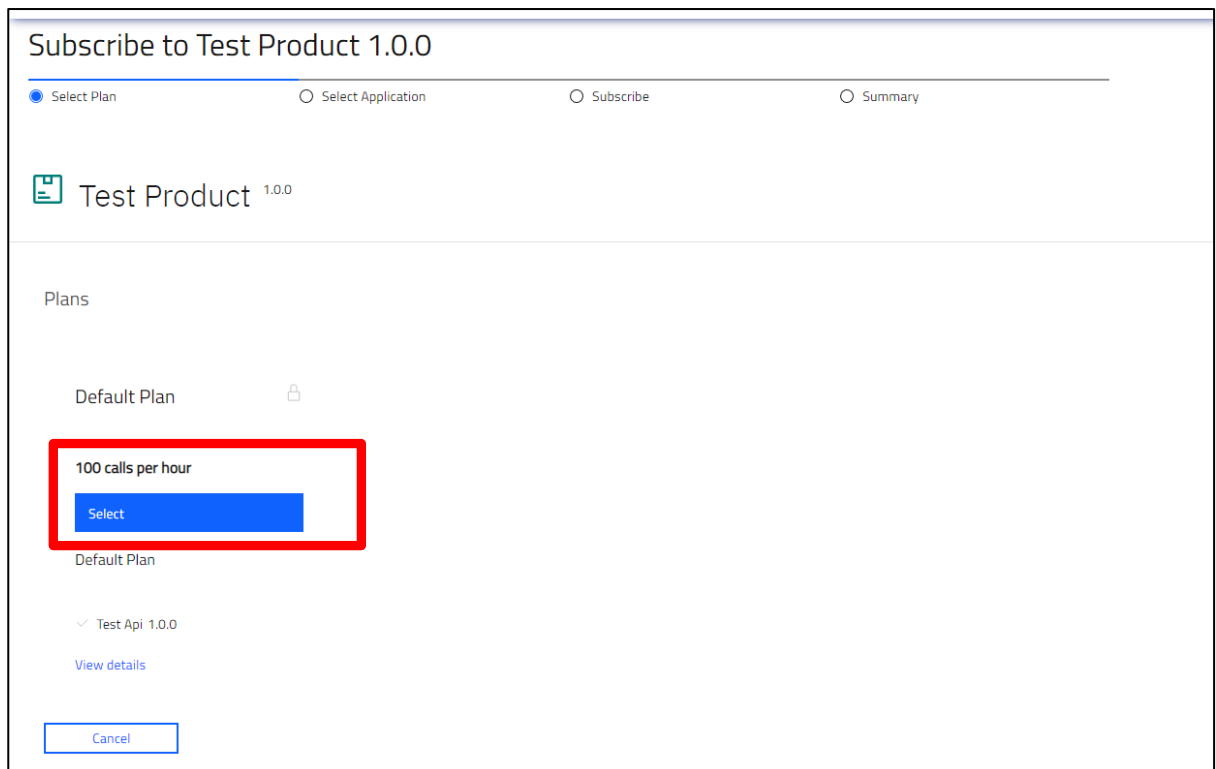
- g) You can search for the available Products and select the desired product for subscription



- h) Once you click on the desired Product you will get page where its asks for Get Access.



- i) After clicking on Get access you have to select the plan used for the API (100calls/hour etc.).



- j) Once the plan is selected, the process will take user to select application section.

Subscribe to Test Product 1.0.0

☒ Select Plan
 ☒ Select Application
 ☐ Subscribe
 ☐ Summary

Select an existing application or create a new application

There is 1 application that is already subscribed to the Test Product product. It is not displayed in this list.

napix_react_app

Test_Overview_API

TEst_APP_NAPIX

+ Create Application

Cancel

Back

5.3 Invoke/Test an API using the Integrated API Console

- a) The user can create a new application by clicking Create Application button. Give an appropriate name for the application (User can give the application name where the API will be consumed).

Create an application

Title *

sqq_app

Description

Application OAuth Redirect URL(s)

Add another item

Save

- b) Once the application gets created the platform will give the credentials for the application
- c) Copy the application credentials (Key and Secret). *Secret will be visible only once. The user needs to keep it with him/her for future use.*

Credentials for your new application

The API Key and Secret have been generated for your application.

Key

.....

Secret

.....

The Secret will only be displayed here one time. Please copy your API Secret and keep it for your records.

- d) Close the application credentials tab after copying the credentials
- e) Select the already created application (application created in Step 12)

Subscribe to Test Product 1.0.0

Select Plan Select Application Subscribe Summary

Select an existing application or create a new application

There is 1 application that is already subscribed to the Test Product product. It is not displayed in this list.

napix_react_app Test_Overview_API Test_APP_NAPIX sqg_app

+ Create Application

Cancel Back

- f) Once the user selects the application, the process will take the user to subscribe section. click on Next to subscribe the API in the created application

Subscribe to Test Product 1.0.0

☒ Select Plan
 ☒ Select Application
 ☒ **Subscribe**
 ☐ Summary

Confirm Subscription

Product	Application	Plan
Test Product	sqg_app	Default Plan: Free subscription for 100 calls per hour

- g) Subscription completes after this process. Now the user will have access to the subscribed API.

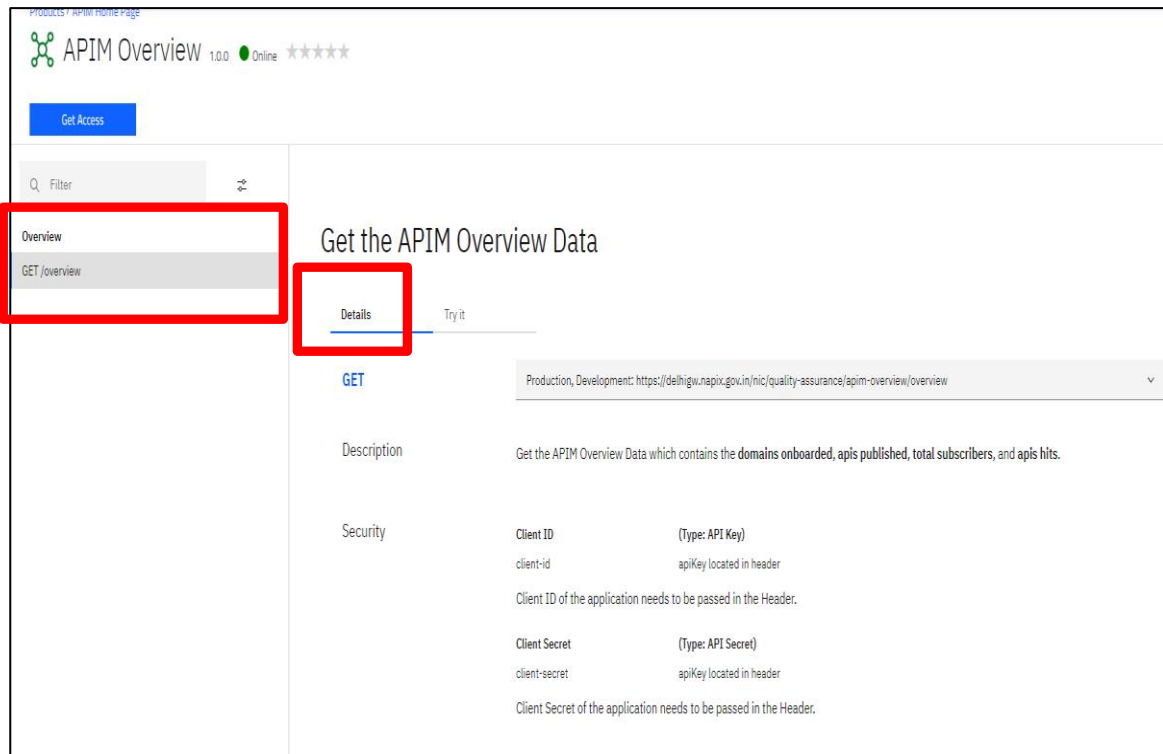
Subscribe to Test Product 1.0.0

☒ Select Plan
 ☒ Select Application
 ☒ Subscribe
 ☒ **Summary**

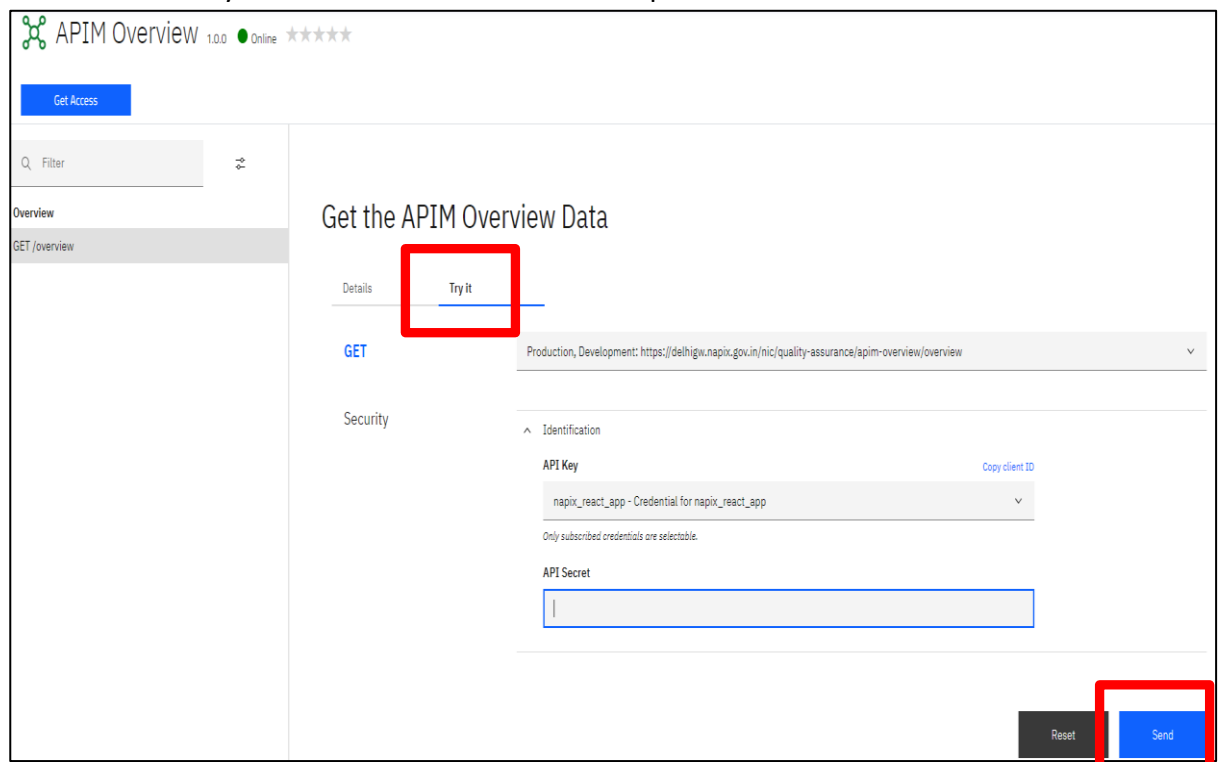
Subscription Complete
 Your subscription request has been created and is now pending approval. You will receive an email once your subscription request is approved.

Product	Application	Plan
Test Product	sqg_app	Default Plan

- h) After clicking done the process will take user to the selected API/Product again.
- i) Click on the product again and it will show the number of APIs available in the Product.
- j) Click on desired API in the product.
- k) It will show all the available methods (GET, POST etc.) in the API (**after overview section**).
- l) User will get the details of API in the Details section.



- m) From the <Details> tab user can find the parameters and URL to be used for calling the API (These details to be used in the application to call the desired API).
- n) User can use <Try it> tab to test the API from the platform also.



5.4 Steps for Consuming the API inside Sample Application

a) Select the API which you have subscribed and want to consume in the application. Go to the Details Tab of the respective API.

Get the APIM Overview Data

Details Try it

GET Production, Development: <https://delhigw.napix.gov.in/nic/quality-assurance/apim-overview/overview>

Description Get the APIM Overview Data which contains the **domains onboarded**, **apis published**, **total subscribers**, and **apis hits**.

Security

Client ID	(Type: API Key)
client-id	apiKey located in header
Client ID of the application needs to be passed in the Header.	
Client Secret	(Type: API Secret)
client-secret	apiKey located in header
Client Secret of the application needs to be passed in the Header.	

Parameters

Header

Accept	Permitted values
optional	application/json

b) Once you click on the API details you will get the HTTP Method Used in the API and Security feature implemented for the API etc

c) The User will also get sample request in curl and various languages like python, java etc. using the API in those languages

Example request

curl

```
curl -X GET https://delhigw.napix.gov.in/nic/quality-assurance/apim-overview/overview \
  -H 'Accept: application/json' \
  -H 'client-id: clientId' \
  -H 'client-secret: clientSecret'
```

Responses

200 200 OK

Body

Example

```
{
  domains_onboarded : 20
  apis_published : 200
  total_subscribers : 200
  total_hits : "20,00,000"
}
```


d) Sample Curl Request for an API

Example request

curl

```
curl --request GET \
--url https://delhigw.napix.gov.in/nic/quality-assurance/apim-overview/overview \
--header 'accept: application/json' \
--header 'client-id: clientId' \
--header 'client-secret: clientSecret'
```

- e) In the above curl request we can see that the URL has been changed to NAPIX gateway URL and in the API client Id and Client Secret is used as security for the API
- f) You can specify the client ID and Client secret received while creating the application (Application through which the API is subscribed) to call the API in your code.

6. Conclusion

The document covered the general idea about Open API management, its lifecycle and how NIC API Exchange Platform can be used for publishing and consuming an API. The Step-by-step guide for Consumer/Developer can be used by any Consumer/Developer for subscribing and consuming an API.

For feedback or suggestions, you may write to support-api@nic.in

Glossary -Definitions and Abbreviations

API: Application Program Interface. An application program interface is a set of routines, protocol, objects or functions that developers use to create the software or interact with the external system. In general it is set of clearly defined methods of communication between various software components.

API Lifecycle Management: Process of managing the life cycle of API. Starting from API creation to API retirement.

API Gateway: Single point of entry for the set of published API's to provide a uniform experience to the user. Gateway is responsible for API request validation like throttling policy validation, token validation etc.

API Publisher: Owner of the API who publishes their API for broader use of the API.

A publisher manages a set of APIs and controls the API lifecycle, subscriptions aspects. The publisher can also control the access patterns for APIs and has access to all API statistics.

API Publisher Portal: Platform for API publishers to create, publish and manage the API lifecycle.

API Subscriber: Subscriber uses the API store to discover APIs, read the API definition and document, obtain access tokens and invoke or test the APIs.

API Developer/Store Portal: Platform for API subscribers to discover, subscribe and test the API.

Consumer: Developer who uses the published/subscribed API in their application.

Discover: Finding the available API's in the Store Portal.

Endpoint: End Point is the URL of a server or service. Each endpoint is the location from which APIs can access the resources they need to carry out in their application.

Get: HTTP method used to request data from a specified resource

HTTP: Hypertext Transfer Protocol .Underlying protocol defines the way in which messages are formatted and transmitted.

IoT: Internet of Things. It is the extension of internet connectivity to the physical objects to allow communication between such devices.

Internal API: Private APIs, or internal APIs, are published internally for use by the developers of the same organization to improve its own services. Private APIs are not exposed to third parties. Available only to NIC applications and users.

JSON: JavaScript Object Notation. Light weight data interchange format in text format. Used for representing simple data structure and objects in web browser.

JWT: Json Web Token. JWT is used for the secured transmission of information between parties in Json object format.

Open API: Public APIs, or open APIs, are published publicly and can be used by any third-party. There are no restrictions on these APIs. Available to non NIC applications and users Examples include the general public, other agencies, or private companies.

OpenAPI Specification (OAS): The OpenAPI Specification (OAS) defines a standard, language-agnostic interface to RESTful APIs which allows both humans and computers to discover and understand the capabilities of the service without access to source .API description format for REST API's.

Partner API: Partner APIs can only be used by specific parties with whom the government agrees to share data. Partner APIs are used within other government entities and partners often to integrate software between the entities.

Post: HTTP method used to send data to create/Update a resource.

Public API: Public APIs, or open APIs, are published publicly and can be used by any third-party. There are no restrictions on these APIs.

REST: Representational State Transfer is an architectural style. The REST architecture allows API providers to deliver data in multiple formats such as plain text, HTML, XML, YAML, and JSON.

SAML2: Security Assertion Mark-up Language. Open standard for sharing security information about identity, authentication and authorization across different systems. It is implemented with the Extensible Mark-up Language XML standard for sharing data.

SOAP: Simple Object Access Protocol. SOAP is a Messaging protocol specification. SOAP can be carried over a variety of lower-level protocols, including the web-related Hypertext Transfer Protocol (HTTP).

Throttling: Process used to control the use of API's by consumers. Administrators and publishers of API manager can use throttling to limit the number of API requests per day/week/month. For example, you can limit the number of total API requests as 10000/day.

TPS: Transactions per Second. In the API Management Platform perspective, it is the number of transactions API can take in one second.

UI: User Interface

URL: Uniform Resource Locator.

Web socket API: Advanced technology which helps in two-way interactive communications between client browser and server. Used in real-time environment where event driven response from the server is also needed.

WSDL: Web Service Description Language. It is a description of web service written in XML format.

References

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- 3) <https://whatistechtarget.com>
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- 5) <https://www.webopedia.com>